

TU Berlin

Civil Systems Engineering

Projekt Systemtechnik SS 25



Project Report

Virtual Temperature Sensing in Existing Buildings: Calibrating EnergyPlus Models with Sparse Sensor Data

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1. Introduction

Buildings consume a large share of global energy, with heating, ventilation, and air conditioning (HVAC) systems as major contributors. Monitoring thermal conditions is critical for comfort, system performance, and energy efficiency, yet dense sensor networks remain impractical in most existing buildings due to cost, disruption, and maintenance. As a result, engineers often have limited data to validate models and improve operations.

Physics-based simulation tools such as EnergyPlus can overcome this gap. When calibrated with measured data, these models act as virtual sensors, extending monitoring beyond physical instrumentation. Seminal studies have confirmed calibration as a foundation of building performance modeling, but they typically assume access to dense sensor networks. In practice, most buildings operate under sparse instrumentation, especially in corridors and auxiliary spaces.

This project addresses that reality. It tests whether EnergyPlus models calibrated with limited sensor data can reproduce thermal conditions with sufficient accuracy to substitute for dense instrumentation. Specifically, it compares a fully calibrated model using ten sensors with a sparsely calibrated model using only three. By quantifying accuracy and cost trade-offs, the study demonstrates the potential of sparse calibration for building operation, renovation, and retrofit planning.

2. Point of Departure

Building energy models simulate thermal behavior and guide design, operation. Tools like EnergyPlus model geometry, systems, and schedules in detail, but their reliability depends on calibration with measured data. Dense sensor networks provide high-quality calibration inputs, yet they are costly, disruptive to install, and often impractical in occupied buildings. As a result, many buildings remain under-instrumented, limiting the ability of engineers to validate and improve model predictions.[1][2]

Seminal contributions demonstrate that calibrated simulation models can produce accurate predictions when abundant data are available. Researchers have refined parameter tuning, optimization algorithms, and error metrics to align simulated outputs with measured conditions. These works established calibration as a cornerstone of building performance modeling. Still, most of the literature assumes well-instrumented environments, focusing on primary spaces such as large offices or meeting rooms where sensors are concentrated. Under-instrumented buildings, which make up most of the global stock, receive far less attention. Efforts on virtual sensors and digital twins point toward new possibilities, but reproducible frameworks for calibration with sparse data remain underdeveloped.[3][4]

2.1. Gap Analysis

Most calibration studies assume dense monitoring. They focus on buildings with extensive sensor networks or on new projects where instrumentation can be designed into the building from the start. This leaves out the reality of most existing buildings, where cost and disruption make dense monitoring impractical.

Research also tends to emphasize instrumented spaces that drive comfort and control, such as offices and meeting rooms, while overlooking corridors, auxiliary rooms, and smaller offices.

These secondary zones influence airflow and thermal propagation across the building. Ignoring them undermines the realism and generalizability of calibrated models.

The trade-off between cost and accuracy also remains poorly understood. Ten sensors provide rich calibration data but at high cost, while three sensors cut cost by 70% yet risk weaker performance. Although this tension is recognized, seminal work rarely provides evidence to guide decisions, leaving facility managers to rely on intuition when balancing instrumentation cost against accuracy.

2.2. *Departure and Positioning*

This research builds on seminal work while addressing its blind spots. It affirms the validity of calibration metrics such as RMSE, CVRMSE, and MBE, but tests them under sparse data conditions. It develops calibration as a form of virtual sensing, but in environments where physical sensors are few. Most importantly, it quantifies the trade-off between cost and accuracy, offering evidence where previous work offered only assumption.

In this way, the study respects the foundations of calibration research while extending it into a domain that earlier contributions identified but did not resolve: the calibration of under-instrumented buildings.

2.3. *Research Question and Hypotheses*

This study examines whether an EnergyPlus model calibrated with sparse temperature sensor data can generate reliable virtual estimates in under-instrumented zones and how its predictive accuracy compares with a fully calibrated model. The contribution lies in explicitly testing the balance between cost efficiency and calibration accuracy, an issue largely overlooked in current research. The central research question guiding this work is whether a physics-based EnergyPlus model, when calibrated with sparse sensor input, can reproduce thermal conditions in both primary and secondary zones with sufficient accuracy to serve as a practical alternative to dense instrumentation.

Three hypotheses structure the analysis. The first assumes that a fully calibrated model using all available sensors delivers the most accurate zone-level temperature predictions and provides the performance benchmark. The second holds that a sparsely calibrated model using only three sensors can still generate virtual data for uncalibrated zones with acceptable accuracy, provided the chosen calibration zones are spatially diverse and representative. The third posits that both strategies can estimate temperatures in unmonitored zones, but sparse calibration offers a cost-efficient alternative, albeit with some reduction in precision relative to full calibration.

3. *Research Methodology*

This study used a reproducible simulation-based methodology combining EnergyPlus for building performance modeling, OpenStudio for model development, and Python for calibration and optimization. The aim was to test whether sparse calibration with limited sensor data could replicate the accuracy of full calibration and to quantify the trade-off between cost and accuracy across primary and secondary zones.

3.1. *Case Study Building*

The case study was a multizone office building consisting of eighteen thermal zones. Ten zones were classified as *primary*, representing regularly occupied spaces such as offices and meeting

rooms where sensors are typically deployed. Eight zones were classified as *secondary*, representing corridors and auxiliary rooms that generally remain un-instrumented. The building geometry and HVAC systems were modeled in OpenStudio, and the baseline input data file (IDF) was generated for EnergyPlus simulation.[5]

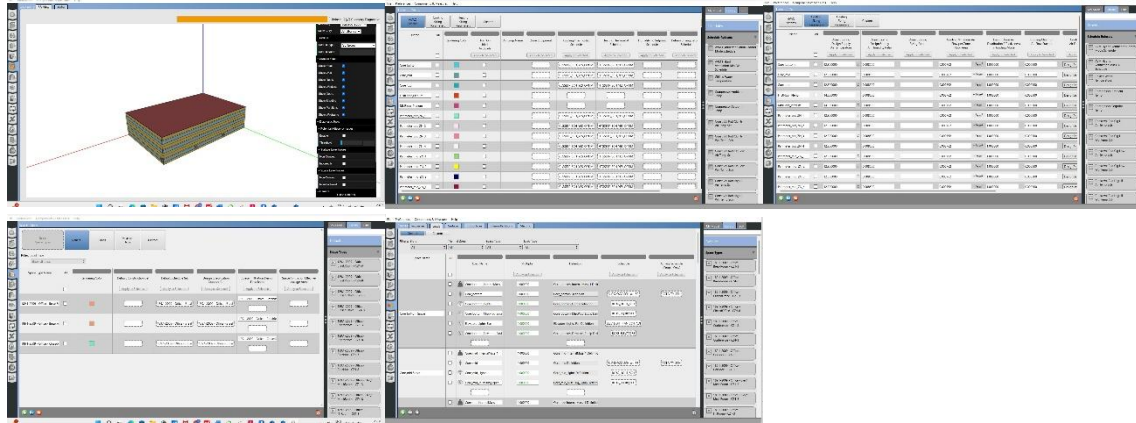


Figure 1. Case study building model in OpenStudio showing geometry, zoning, schedules, and HVAC configuration inputs used for EnergyPlus simulation.

3.2. Synthetic Reference Data

Because measured sensor data were unavailable, the reference dataset was synthetically generated by running the baseline EnergyPlus model under fixed assumptions. This provided a consistent, noise-free benchmark for comparing calibration strategies. Although synthetic data lack the noise, drift, and gaps of real measurements, they allowed controlled experimentation and isolated the effect of calibration methods on accuracy. The dataset therefore served as the reference for evaluating outcomes.

3.3. Calibration Parameters

Calibration targeted six operational parameters:

1. Internal gains multiplier
2. Infiltration multiplier
3. VAV minimum airflow multiplier
4. Reheat coil capacity multiplier
5. Minimum outdoor air flow multiplier
6. Supply air temperature offset

These parameters were chosen because they directly influence thermal dynamics and can absorb uncertainty in physical characteristics that are difficult to measure in practice, such as insulation degradation, air leakage, or undocumented retrofits.

3.4. Calibration and Optimization Workflow

Calibration was automated in Python through a three-step loop:

- **Parameter modification.** Python scripts modified the IDF file by scaling the six calibration parameters within defined ranges.
- **Simulation execution.** The modified model was simulated with EnergyPlus, producing hourly zone-level temperature outputs.

- **Performance evaluation.** Root mean square error (RMSE) was calculated between simulated and reference temperatures, serving as the calibration objective function.

The optimization algorithm was differential evolution, a global stochastic optimizer in SciPy. It was chosen for its ability to handle non-linear, multi-parameter spaces and avoid local minima that trap gradient-based methods. Parameter bounds reflected realistic conditions—for example, infiltration multipliers between 0.5–1.5 and supply air temperature offset between $-1.0\text{ }^{\circ}\text{C}$ and $+1.0\text{ }^{\circ}\text{C}$. Each iteration generated a parameter set, executed a simulation, and evaluated RMSE until convergence criteria or iteration limits were reached.

3.5. Calibration Strategies

Two calibration strategies were compared, differing only in the number of zones used as anchors.

Full Calibration (10 Sensors)

This strategy used reference data from all ten primary zones. By minimizing RMSE across main occupied spaces, the resulting model represented the most accurate version under the setup and served as the benchmark.

Sparse Calibration (3 Sensors)

This strategy used only three of the ten primary zones, a 70% reduction in sensor input. Zones were chosen for spatial and thermal diversity and low baseline RMSE in the fully calibrated model, ensuring representative coverage. After calibration, the model generated temperature predictions for the remaining zones, treated as “virtual sensor” outputs.

3.6. Simulation of Un-instrumented Zones

After calibration, both models simulated temperatures in eight secondary zones without sensors, including corridors and auxiliary rooms that influence building dynamics but are rarely instrumented. Outputs were treated as virtual sensor readings and evaluated against the synthetic reference dataset using RMSE. This analysis tested the stability and accuracy of virtual estimates under both calibration strategies and offered insight into their applicability for real-time monitoring in under-instrumented areas.

4. Results

4.1. Strategy 1 (Full Calibration)

The model was calibrated using data from 10 instrumented zones. The calibration achieved a best-fit RMSE of $0.387\text{ }^{\circ}\text{C}$ across these zones. Representative time-series plots for the calibrated zones are shown in Figure 2, illustrating the alignment between the calibrated model, the measured synthetic data, and the un-calibrated base model for each zone. The best-fit calibration parameters obtained for this configuration are summarized in Table 1, presented alongside the parameters derived from Strategy 2 for direct comparison.

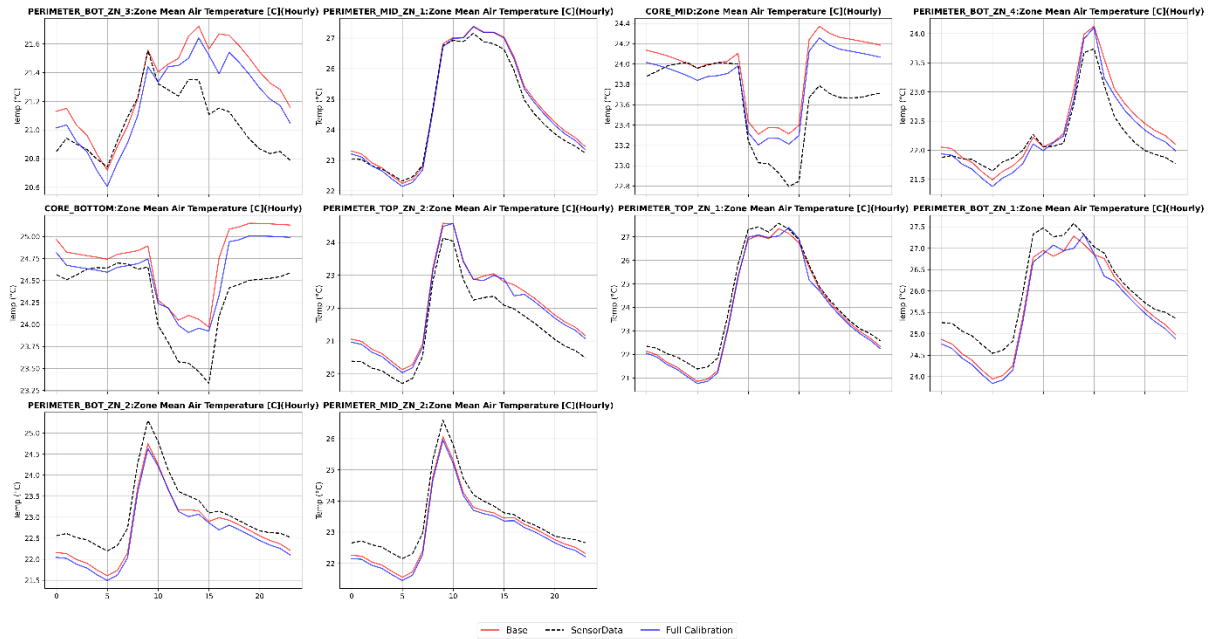


Figure 2. Comparison of zone air temperatures for Strategy 1 (Full Calibration) using the base, measured, and calibrated data.

Table 1. Best-fit calibration parameters for Strategy 1 and Strategy 2.

Parameter	Strategy 1	Strategy 2
Internal Gains Multiplier	0.978	0.912
Infiltration Multiplier	0.767	1.181
VAV Minimum Flow Multiplier	0.255	0.335
Reheat Capacity Multiplier	0.980	1.005
Outdoor Air Minimum Flow Multiplier	0.523	1.057
Supply Air Temperature Offset	-0.888	-0.157

4.2. Strategy 2 (Sparse Calibration)

The model was calibrated using data from 3 of the instrumented zones. The calibration achieved a best-fit RMSE of 0.22 °C across these zones. Representative time-series plots are shown in Figure 3, comparing the calibrated outputs with the measured synthetic data and the base model.

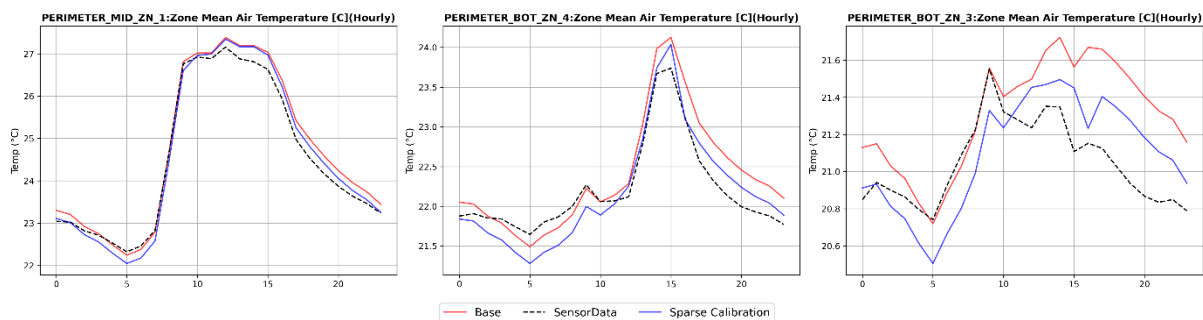


Figure 3. Comparison of zone air temperatures for Strategy 2 (Sparse Calibration) using the base, measured, and calibrated data.

The remaining 7 instrumented zones were not included in this calibration and were simulated using the calibrated model, effectively acting as virtual temperature sensors. For these zones, RMSE values were computed by comparing the simulated results with the available synthetic measurements, and representative time-series plots are shown in Figure 4.

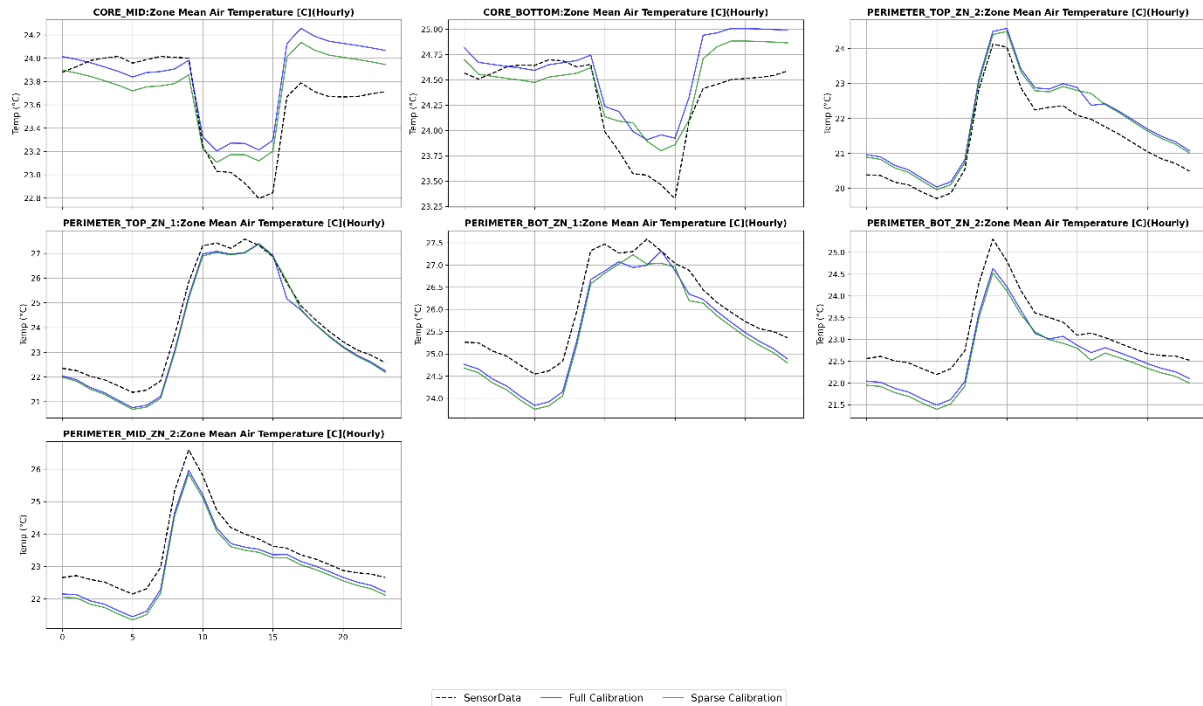


Figure 4. Comparison of zone air temperatures across the remaining 7 zones in Strategy 2 (Sparse Calibration), showing virtual data against the base model and full calibrated model.

A summary comparison between the fully calibrated model (Strategy 1) and the sparsely calibrated model with the 7 virtual zones (Strategy 2) is presented in Table 2, including the percentage differences. This table consolidates the RMSE values from both strategies and provides a direct quantitative basis for assessing calibration accuracy under varying levels of sensor availability.

Table 2. RMSE values for Strategy 1 (Full Calibration) and Strategy 2 (Sparse Calibration with 3 calibration and 7 validation zones).

Zone	RMSE (°C) – Strategy 1 (Full Calibration)	Zone	RMSE (°C) – Strategy 2 (Sparse Calibration)	% Difference
Zone 1	0.230	Zone 1	0.215	–6.7 %
Zone 2	0.237	Zone 2	0.230	–3.0 %
Zone 3	0.259	Zone 3	0.216	–16.6 %
Zone 4	0.307	Virtual Zone 1	0.252	–17.7 %
Zone 5	0.348	Virtual Zone 2	0.276	–20.7 %
Zone 6	0.539	Virtual Zone 3	0.492	–8.8 %
Zone 7	0.422	Virtual Zone 4	0.452	+7.2 %

Zone 8	0.504	Virtual Zone 5	0.575	+14.1 %
Zone 9	0.514	Virtual Zone 6	0.606	+17.8 %
Zone 10	0.505	Virtual Zone 7	0.601	+19.1 %
Average	0.387	Average	0.392	+1.3 %

4.3. Un-instrumented Zones

For the 8 un-instrumented zones, both calibration strategies generated virtual temperature profiles. As no sensor measurements were available, RMSE values were computed relative to the un-calibrated base model. The results are presented in Table 3, and representative time-series profiles are shown in Figure 5, comparing the outputs from both strategies with the base model.

For un-instrumented zones, errors were computed relative to the un-calibrated base model due to the absence of measured data. This reliance on synthetic reference values represents a methodological limitation and helps explain the higher RMSE observed under sparse calibration, as the model lacks ground-truth anchors in these secondary spaces.

Table 3. RMSE values for Strategy 1 (Full Calibration) and Strategy 2 (Sparse Calibration) for un-instrumented zones.

Zone	Strategy 1 (Full Calibration)	Strategy 2 (Sparse Calibration)	% Difference
Zone 11	0.090	0.148	+64.0 %
Zone 12	0.102	0.186	+82.4 %
Zone 13	0.119	0.215	+80.7 %
Zone 14	0.143	0.176	+23.6 %
Zone 15	0.111	0.157	+41.4 %
Zone 16	0.120	0.153	+27.5 %
Zone 17	0.104	0.207	+99.4 %
Zone 18	0.102	0.204	+100.0 %
Average	0.111	0.181	+63.1 %

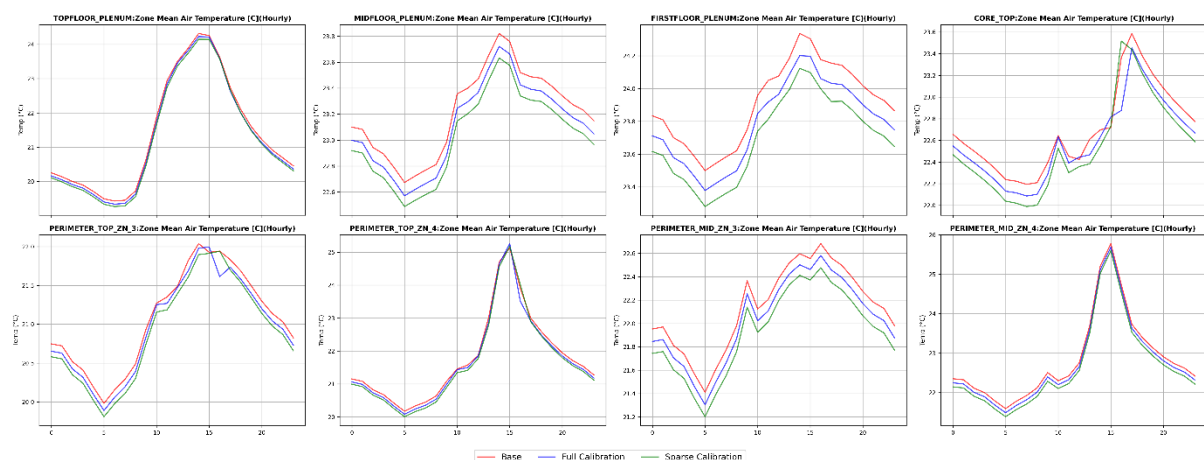


Figure 5. Comparison of zone air temperatures across the un-instrumented zones under the base model, the fully calibrated model (Strategy 1), and the sparsely calibrated model (Strategy 2)

5. Discussion

5.1. Contribution

This research establishes a reproducible methodology for using EnergyPlus as a virtual sensing framework in buildings with limited instrumentation. By comparing full and sparse calibration strategies, it provides empirical evidence of the cost–accuracy trade-off in building performance modeling and demonstrates how the same approach can guide renovation and retrofit planning by evaluating future interventions with limited sensor data instead of dense instrumentation.

The study shows that sparse calibration can replicate the accuracy of full calibration in instrumented zones, with an average difference of only 1.3%. In fact, sparse calibration outperformed full calibration in six of ten zones, including Zone 5, where the RMSE dropped from 0.348 °C to 0.276 °C (a 20.7% improvement). This finding challenges the assumption that dense sensor networks are always necessary and proves that carefully chosen calibration zones can deliver comparable results at much lower cost. At the same time, the results clarify the limits of sparse calibration: in secondary, un-instrumented zones, errors increased by more than 60%, with RMSE nearly doubling in Zone 18 (0.102 °C to 0.204 °C). This dual insight—strengths and boundaries—makes the contribution both practical and realistic.

The study also quantifies the trade-off between cost and accuracy. Reducing the number of sensors by 70% (from ten to three) produced almost no loss of accuracy in instrumented zones, with average RMSE increasing only from 0.387 °C to 0.392 °C (+1.3%). In secondary zones, however, the same reduction raised the average RMSE from 0.111 °C to 0.181 °C (+63.1%), a penalty that highlights the limits of sparse calibration. This trade-off illustrates a practical decision point: the marginal savings from installing fewer sensors may be justified in primary, instrumented areas where accuracy remains high, but the same reduction creates unacceptable uncertainty in secondary spaces. By quantifying both the efficiency gained and the precision lost, the study equips facility managers and engineers with data-driven guidance for deciding when sparse calibration is a viable cost-effective strategy and when additional instrumentation is essential. This trade-off is illustrated in Figure 6, which compares full and sparse calibration across cost and accuracy dimensions, making clear that sparse calibration preserves accuracy in primary zones while imposing significant penalties in secondary ones.

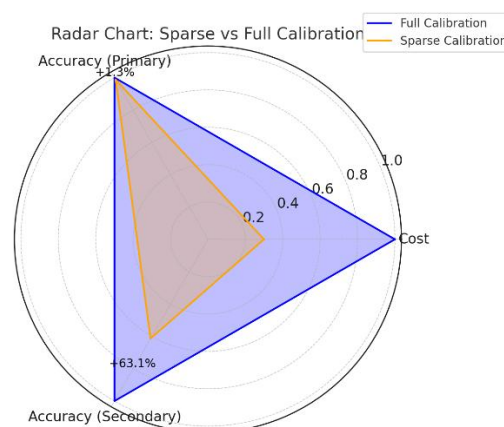


Figure 6. Radar chart comparing sparse and full calibration across cost and accuracy dimensions

Finally, the research extends the role of calibrated models beyond monitoring. Once aligned with real building behaviour, the model becomes a tool for testing design alternatives such as insulation upgrades, glazing improvements, or HVAC interventions. This positions calibration not only as a way to reproduce present conditions but also as a foundation for evidence-based renovation and retrofit planning.

5.2. Limitations

This study has several limitations. It relied on case-specific synthetic data, which enabled controlled experimentation but lacked the noise, drift, and gaps of real sensor streams. As a result, RMSE values reported here are likely optimistic, whereas calibration against real measurements typically yields higher errors.[6][7]

Calibration also considered only zone air temperature. While temperature is a key indicator of comfort and HVAC performance, other variables—humidity, CO₂, energy use, and occupancy—also shape building dynamics, and their exclusion narrows the scope.

The method further assumed that building properties matched documented specifications. In reality, insulation degrades, seals leak, and equipment loses efficiency. Undocumented retrofits may also alter systems. By not accounting for these effects, the model absorbed uncertainty into operational multipliers, preserving accuracy in sensor-equipped zones but reducing realism elsewhere.

Another limitation concerns secondary, un-instrumented zones. Because no measured data were available, errors were computed relative to the un-calibrated base model. This reliance on synthetic reference values likely amplified RMSE differences and reduces generalizability.

Finally, the optimization minimized only RMSE. Other metrics such as CVRMSE or MBE might have produced different outcomes. The parameter space was also limited to operational multipliers. Expanding it to physical properties could improve accuracy but at higher computational cost. Balancing accuracy and efficiency remains a challenge for future work.

5.3. Future Research

Future work should test this methodology in real buildings with real sensor data. This study relied on synthetic data, which simplified calibration and produced optimistic results. Applying the approach to noisy, incomplete, or drifting datasets would test robustness and expose practical challenges.

Research should also extend beyond single-variable calibration. While air temperature proved useful, performance depends on humidity, CO₂, energy use, and occupancy. Incorporating multiple streams would create a more comprehensive framework capturing both comfort and efficiency.

Improving accuracy in secondary zones is another priority. Sparse calibration performed well in primary areas but lost precision in corridors and auxiliary rooms. This gap could be addressed by combining simulation with machine learning or optimizing sensor placement.

Models should also be validated for decision support. Comparing simulated retrofit scenarios—such as insulation, glazing, or HVAC upgrades—with actual outcomes would test predictive reliability and strengthen their role in planning.

Finally, two directions expand scope. Inverse Performance Modeling (IPM) could use discrepancies between documented and calibrated parameters to detect deterioration or undocumented retrofits. Dynamic (online) calibration would continuously align digital twins with live data, adapting to seasonal changes, occupancy shifts, and equipment wear. Both would turn calibration into an ongoing intelligence source for building operation and maintenance.

5.4. Impact

This study delivers both academic and practical impact. By showing that sparse calibration can replicate full calibration in instrumented zones with only a 1–2% accuracy loss, it demonstrates that reliable monitoring is possible with far fewer sensors. This reduces installation and maintenance costs while avoiding disruption from dense instrumentation, benefiting owners and operators. Occupants gain from improved thermal control, though secondary zones may show inconsistencies.

The method also supports environmental goals. HVAC systems account for a large share of building energy use, and inaccurate models often drive waste. By improving accuracy in monitored zones, sparse calibration enables more efficient heating, cooling, and ventilation. Dynamic calibration extends this benefit by adapting models to seasonal shifts, occupancy, and equipment wear, helping organizations meet carbon-reduction targets. Policymakers also benefit, as these tools enable compliance with energy codes and sustainability benchmarks. The trade-off is computational complexity: continuous calibration requires more processing and data handling, which may challenge smaller operators.

Economically, facility managers save by reducing sensor deployment—three sensors achieved nearly identical results to ten, cutting instrumentation by 70%. Investors gain confidence by testing retrofit measures virtually before committing capital. Service providers, however, may see reduced revenue from hardware sales even as demand grows for calibration and analytics services.

Altogether, the study shows how a calibrated building model can extend monitoring in under-instrumented buildings, support evidence-based retrofit planning, and improve management. The approach impacts stakeholders across the spectrum: owners and operators gain cost-effective monitoring tools; occupants benefit from comfort and safety; regulators see stronger compliance pathways; and service providers face both new opportunities and challenges. These shifts mark a move toward smarter, more efficient, and sustainable operations.

6. Conclusion

This study demonstrates that EnergyPlus models can be effectively calibrated with sparse sensor data to generate reliable thermal predictions in primary zones while reducing instrumentation needs by 70%. The sparse calibration strategy matched the accuracy of full calibration in instrumented areas, with average RMSE increasing only from 0.387 °C to 0.392 °C (+1.3%). In some cases, sparse calibration even outperformed full calibration, proving that

carefully chosen calibration zones can anchor robust models at lower cost. However, accuracy decreased substantially in secondary, un-instrumented spaces, with RMSE rising by more than 60%. This dual outcome underscores both the promise and the limits of sparse calibration.

The contributions are twofold. First, the research delivers a reproducible methodology that blends simulation, calibration, and optimization into a framework for virtual sensing under sparse data conditions. Second, it quantifies the cost–accuracy trade-off, equipping stakeholders with evidence-based guidance for deciding where sparse calibration is a viable alternative and where additional instrumentation is essential.

Limitations remain, including the use of synthetic data and the exclusive focus on temperature. Future work should test the method with real sensor streams, extend calibration to other performance variables, and explore advanced approaches such as inverse performance modeling and dynamic (online) calibration.

Ultimately, the study shows that sparse calibration is not only feasible but also valuable. It offers building owners, operators, and policymakers a practical pathway to reduce monitoring costs while still enabling effective management, renovation planning, and sustainability improvements. By balancing efficiency with precision, this approach advances the broader goal of making existing buildings smarter, more resilient, and more sustainable.

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